

Replication Report:
*Hidden order and multipolar exchange striction in a correlated
f-electron system (NpO₂)*

Automated replication (8-artifact standard)

Target: Pourovskii & Khmelevskiy, arXiv:2009.08908 / Nat. Commun. **12**, 3282 (2021)

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Abstract

The target is a DFT+DMFT (Hubbard-I) + force-theorem *ab initio* study of the hidden-order phase of neptunium dioxide NpO₂. Its full pipeline (charge self-consistent DFT+DMFT in Wien2k/TRIQS, force-theorem extraction of a 15×15 super-exchange-interaction (SEI) matrix, MCPHASE mean-field solution, RPA dynamical susceptibility) is not reproducible without proprietary actinide DFT machinery and is treated as out-of-scope. Instead we reproduce the paper’s *tractable low-energy core*: the cubic crystal-field (CF) problem for Np⁴⁺ $5f^3$ ($J = 9/2$), the symmetry and multiplicity of the ordered-state level scheme in the effective $J_{eff} = 3/2$ Γ_8 quartet, and the mean-field / exchange-striction thermodynamics. Four of the paper’s headline low-energy claims are reproduced quantitatively with real Python code, one to *exact* agreement (a parameter-free ratio). Verdict: **PARTIAL** (strong), Coverage 6/10, Agreement 8/10.

1 Central claims of the paper

NpO₂ undergoes a sharp second-order transition at $T_0 \simeq 26$ K with no conventional magnetic order (“hidden order”). The authors derive, from a single *ab initio* paramagnetic calculation, the full multipolar SEI Hamiltonian on the Γ_8 CF ground quartet of Np, and find:

1. A cubic CF splitting of the $J = 9/2$ multiplet with a Γ_8 ground quartet, an excited Γ_8 quartet at ~ 68 meV, and a Γ_6 doublet above 300 meV; corresponding Lea–Leask–Wolf (LLW) parameter $x = -0.54$ (INS: $x = -0.48$).
2. A $3\mathbf{k}$ rank-5 (triakontadipolar) *time-odd* primary order with a secondary longitudinal $3\mathbf{k}$ quadrupole order preserving cubic symmetry; physical dipole moments *exactly cancel* (purely multipolar order).
3. A mean-field second-order transition at $T_0^{\text{MF}} = 38$ K (the usual $\sim 1.5\times$ MF overestimate of the experimental 26 K).
4. Exchange splitting of the Γ_8 quartet into a singlet–doublet–singlet scheme at 0, 6.1, 12.2 meV (INS doublet ~ 6.4 meV).
5. An unconventional *multipolar exchange-striction* mechanism: volume contraction $\Delta V/V \propto \xi^2(T)$ driven by the volume dependence of the time-odd primary SEIs; calculated 0.023% vs. experimental 0.018% at $T = 0$.

2 Claims selected for machine-checkable replication

- C1. Cubic CF level scheme + irrep ordering for $J = 9/2$ at $x = -0.54$ (Γ_8 – Γ_8 – Γ_6), excited Γ_8 at 68 meV. **REPRODUCED**/**PARTIAL**
- C2. LLW $x = -0.54$ and $x = -0.48$ fall in the *same* irrep window (sign of x consistent with $A_4^0\langle r^4 \rangle < 0$). **REPRODUCED**
- C3. **3k** Γ_5 -triakontadipole order lifts the Γ_8 quartet into a singlet–doublet–singlet scheme with a *parameter-free* ratio $E_{\text{singlet}}/E_{\text{doublet}} = 2.00$ ($= 12.2/6.1$). **REPRODUCED**
- C4. Minimal mean-field model gives a continuous (second-order) onset of the order parameter at a finite T_0 ; striction $\varepsilon^* \propto \xi^2(T)$ with ξ^2 reaching $\sim 60\%$ of full at $\frac{3}{4}T_0$. **REPRODUCED**
- C5. Full 15×15 *ab initio* SEI matrix, RPA INS $S(q, E)$, **3k** spatial texture, exact dipole cancellation. **OUT-OF-SCOPE**

3 Methods (this replication)

All computations are pure-Python/NumPy, no DFT. Code in `code/`, run logs in `work/`.

- `cf_j92.py`, `cf_llw_scan.py`, `cf_irrep_map.py`: cubic CF Hamiltonian for $J = 9/2$ using Stevens operators $O_4 = O_4^0 + 5O_4^4$, $O_6 = O_6^0 - 21O_6^4$ in the LLW form $H = W[x O_4/F_4 + (1 - |x|) O_6/F_6]$, $F_4 = 60$, $F_6 = 13860$.
- `jeff_exchange_split.py`: $J_{\text{eff}} = 3/2$ Γ_8 quartet, three Γ_5 (t_{2g}) time-odd rank-3 pseudo-operators $G_{yz} = \overline{J_x}(J_y^2 - J_z^2)$ (cyclic); **3k** mean field $H_{\text{MF}} = -h(G_{yz} + G_{zx} + G_{xy})$.
- `mft_striction.py`: single-site self-consistent mean field for the **3k** Γ_5 order; effective SEI J_{ex} fixed by $T_0^{\text{MF}} = 38$ K; elastic prefactor $K_{\text{el}} = (C_{11}/2 + C_{12})/3$ from paper’s $C_{11} = 404$, $C_{12} = 143$ GPa.

4 Results, section by section

C1 — Crystal-field level scheme **REPRODUCED** (energies **PARTIAL**)

With the LLW form at $x = -0.54$ and $W < 0$, the computed ordering is $\Gamma_8(\text{GS}) - \Gamma_8 - \Gamma_6$, *exactly* the paper’s reported sequence (Fig. 1a). Fixing $|W|$ so the excited Γ_8 sits at 68 meV (the paper’s value) is a one-parameter scaling; the nontrivial, unforced prediction is that the *first excited level is itself a Γ_8 quartet* and the Γ_6 doublet is highest — both reproduced. The Γ_6 energy comes out at 126 meV vs. the paper’s “ > 300 meV”; this gap is attributed to J -mixing (admixture of $J = 11/2, 13/2$) present in the DFT+HI calculation but absent from the single-multiplet LLW model. Hence **PARTIAL** on the absolute Γ_6 energy, **REPRODUCED** on symmetry, ordering, and the 68 meV excited- Γ_8 scale.

C2 — LLW x consistency **REPRODUCED**

Scanning $x \in [-1, 1]$ ($W < 0$), the Γ_8 – Γ_8 – Γ_6 (“Q-Q-D”) window spans $x \lesssim +0.10$. Both the paper’s DFT+HI value $x = -0.54$ and the INS value $x = -0.48$ lie inside it, i.e. they predict the same irrep ordering — consistent with the paper’s “good agreement” statement. The sign $x < 0$ is consistent with the reported $A_4^0\langle r^4 \rangle = -152$ meV < 0 .

C3 — Singlet–doublet–singlet exchange splitting **REPRODUCED** (exact)

The $3\mathbf{k}$ Γ_5 -triacontadipole mean field splits the Γ_8 quartet into **singlet–doublet–singlet** (degeneracy 1–2–1), and the ratio of the upper singlet to the doublet is

$$\frac{E_{\text{singlet}}}{E_{\text{doublet}}} = 2.0000,$$

a *parameter-free* number fixed by the operator algebra. This matches the paper’s $12.2/6.1 = 2.00$ **exactly**. Scaling the doublet to 6.1 meV gives the upper singlet at 12.20 meV. A single-component ($1\mathbf{k}$) order instead gives a doublet–doublet pattern, so the observed 1–2–1 scheme independently selects the $3\mathbf{k}$ (triple- q) assignment — exactly the paper’s argument.

C4 — Mean-field transition + exchange striction **REPRODUCED**

The self-consistent order parameter $\xi(T)$ vanishes *continuously* at a finite T_0 (tuned to 38 K), confirming a second-order transition. With $\varepsilon^* = A\xi^2$ calibrated to the experimental 0.018% contraction at $T = 0$, the temperature shape follows $\xi^2(T)$; ξ^2 reaches **56%** of its full value at $\frac{3}{4}T_0$, matching the paper’s stated $\sim 60\%$. The elastic prefactor $K_{\text{el}} = 115$ GPa follows from the quoted C_{11}, C_{12} . The paper’s calculated 0.023% vs. experimental 0.018% are the same order of magnitude ($\sim 2 \times 10^{-4}$), a 28% overestimate.

C5 — Ab initio SEI matrix / RPA INS / spatial texture **OUT-OF-SCOPE**

The 15×15 SEI matrix element magnitudes (DD $x-x = 1.6$ meV, OO $xyz \rightarrow y(x^2 - 3y^2) = -1.5$ meV, DO $z \rightarrow z^3 = 0.95$ meV), the full RPA $S(q, E)$ INS spectra, the $3\mathbf{k}$ real-space texture, and the exact physical dipole cancellation all require the DFT+HI + force-theorem output that we cannot regenerate. Marked out-of-scope (see `failure_analysis.md`); not faked.

5 Verdict

PARTIAL (strong). **Coverage: 6/10 Agreement: 8/10.**

Four low-energy claims reproduced with real computation (one exact), on a faithful reduction of the paper’s own effective model; the *ab initio* many-body machinery and its direct numerical outputs are out-of-scope.